

ZOOM

inspect network traffic
when using the ZOOM app

**COURSE: ADVANCED INTERNET
TECHNOLOGIES**

Professor: Trevisan Martino

Project carried out by : Di Santo Alessandro



INTRODUCTION

Zoom is a **digital communication platform** that allows you to make video conferences, audio calls, chats and webinars in real time.

It was developed by the **Zoom Video Communications** company in 2011 and has experienced exponential growth especially since 2020, becoming one of the most used tools globally for work, teaching and personal remote interactions.

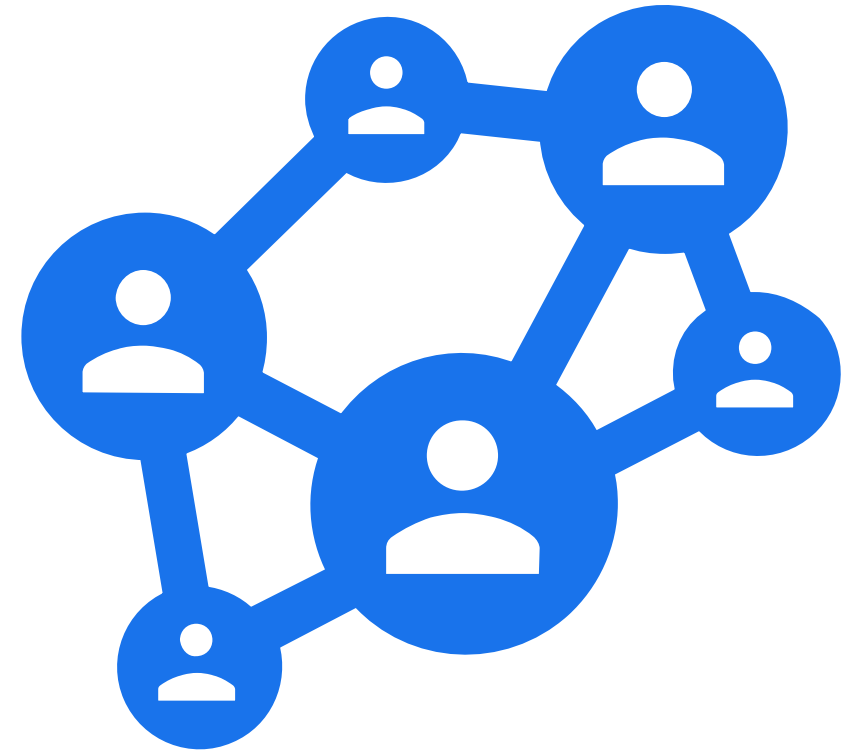
BENEFITS AND USE CASES

Advantages

- **Ease of use:** intuitive interface, quick access via links.
- **Scalability:** suitable for both small meetings and large events.
- **Versatility:** useful for companies, schools, universities, public and private bodies.
- **Accessibility:** doesn't necessarily require an account to join a meeting.

Contexts for use

- **Distance learning:** online lessons, exams, virtual classrooms.
- **Business and work:** team meetings, talks, international conferences.
- **Events and webinars:** product presentations, public conferences, workshops.
- **Personal use:** calls between friends and family, virtual parties, remote meetings



WIRESHARK



We used **Wireshark** to capture our PC traffic while using Zoom.



Wireshark is a **network protocol analyzer**: an opensource tool that allows you to capture and inspect in detail data packets passing through a network.

APP OPENING PHASE

During the **opening phase** of the Zoom app, our PC will perform the following actions:

- The PC queries its **Default Name Server**
- That server will provide it with the public IP of the Zoom server to contact
- Then the PC establishes a **TCP connection** with that server

	Time	Source	Destination	Protocol	Length	Info
23	23:15:24,871409	192.168.1.12	192.168.1.1	DNS	75	Standard query 0x5ac7 A us05web.zoom.us
24	23:15:24,888856	192.168.1.1	192.168.1.12	DNS	91	Standard query response 0x5ac7 A us05web.zoom.us A 170.114.52.5
25	23:15:24,904299	192.168.1.12	us05web.zoom.us	TCP	66	49812 → https(443) [SYN] Seq=0 Win=64240 Len=0 MSS=1460 WS=256 SACK_PERM
26	23:15:24,919385	us05web.zoom.us	192.168.1.12	TCP	66	https(443) → 49812 [SYN, ACK] Seq=0 Ack=1 Win=65535 Len=0 MSS=1452 SACK_PERM WS=8192
27	23:15:24,919526	192.168.1.12	us05web.zoom.us	TCP	54	49812 → https(443) [ACK] Seq=1 Ack=1 Win=66560 Len=0
28	23:15:24,921200	192.168.1.12	us05web.zoom.us	TLSv1.3	571	Client Hello (SNI=us05web.zoom.us)

AUTHENTICATION

- During the zoom **authentication phase** it gives us the ability to authenticate with Google credentials: exactly what we did
- In fact, we note from the figure below a massive exchange of packets with the google server used for authentication: **accounts.google.com**

tls && ip.addr == 192.168.1.12

Wireshark · Conversations · zoom_wire_shark.pcapng

Conversation Settings

☒ Risoluzione dei nomi

☐ Ora iniziale assoluta

☒ Limita al filtro di visualizzazio

Ethernet · 1

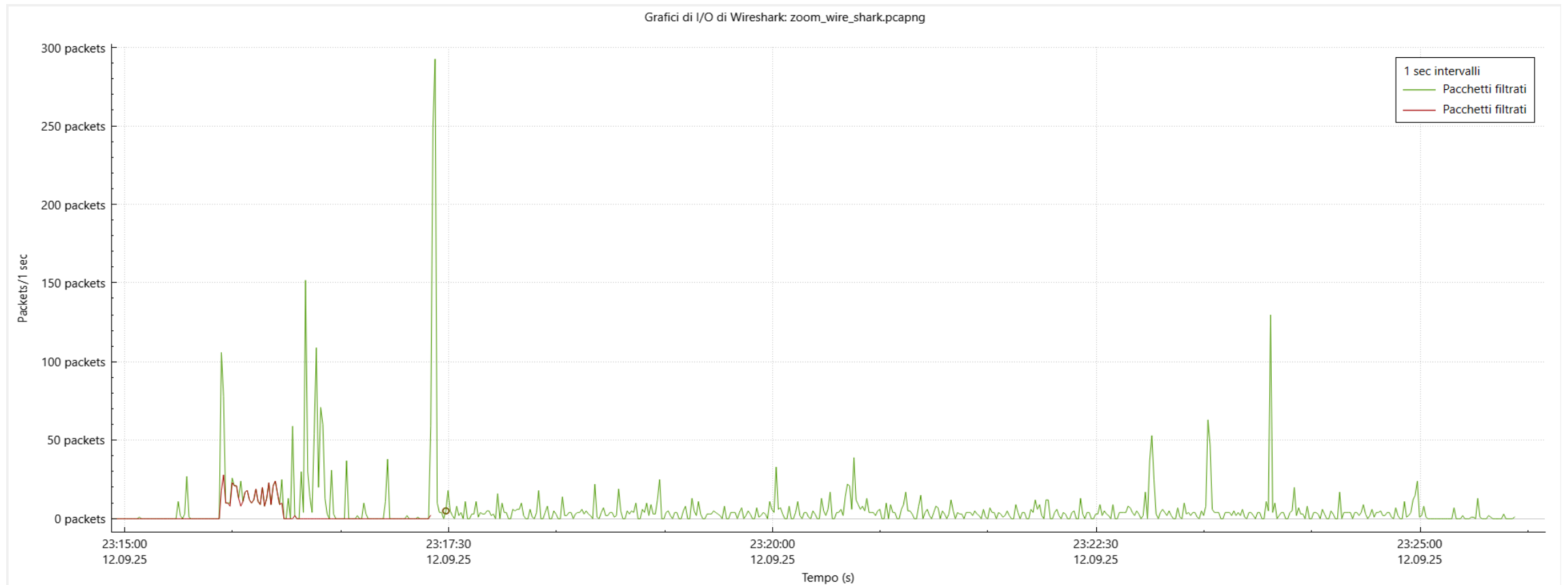
IPv4 · 76

IPv6

TCP · 133

UDP · 35

Indirizzo A	Indirizzo B	Pacchetti	Byte	ID flusso	Pacchetti totali	Percentuale filtrati	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Inizio Rel	Durata	Bits/s A → B
192.168.1.12	accounts.google.com	436	675 kB	18	1.424	30.62%	23	7 kB	413	668 kB	51.701196	96.6367	578 bits/s
192.168.1.12	zoomdca42mmr.dca.zoom.us	1.927	395 kB	61	10.890	17.70%	1.196	297 kB	731	98 kB	148.699919	456.4362	5212 bits/s
192.168.1.12	us05web.zoom.us	199	251 kB	6	680	29.26%	50	20 kB	149	231 kB	31.324341	635.2532	248 bits/s



I/O GRAPHIC

- TLS traffic between **my PC** and the **Google server** used for authentication, in red
- TLS traffic between **my PC** and a **Zoom server** (206.247.42.42), in green

AUDIO/VIDEO COMMUNICATIONS

```
192.168.1.12      zoomdca42mmr.dca.... UDP      123 50205 → 8801 Len=81
zoomdca42mmr.dca.... 192.168.1.12      UDP      133 8801 → 50205 Len=91
192.168.1.12      zoomdca42mmr.dca.... WireGu... 151 Transport Data, receiver=0x2F610F0C, counter=1873780912632365248, datalen=77
```

01

Packets with **Wireguard Protocol** are Zoom packages encrypted with **DTLS-SRTP**, for the following reasons:

02

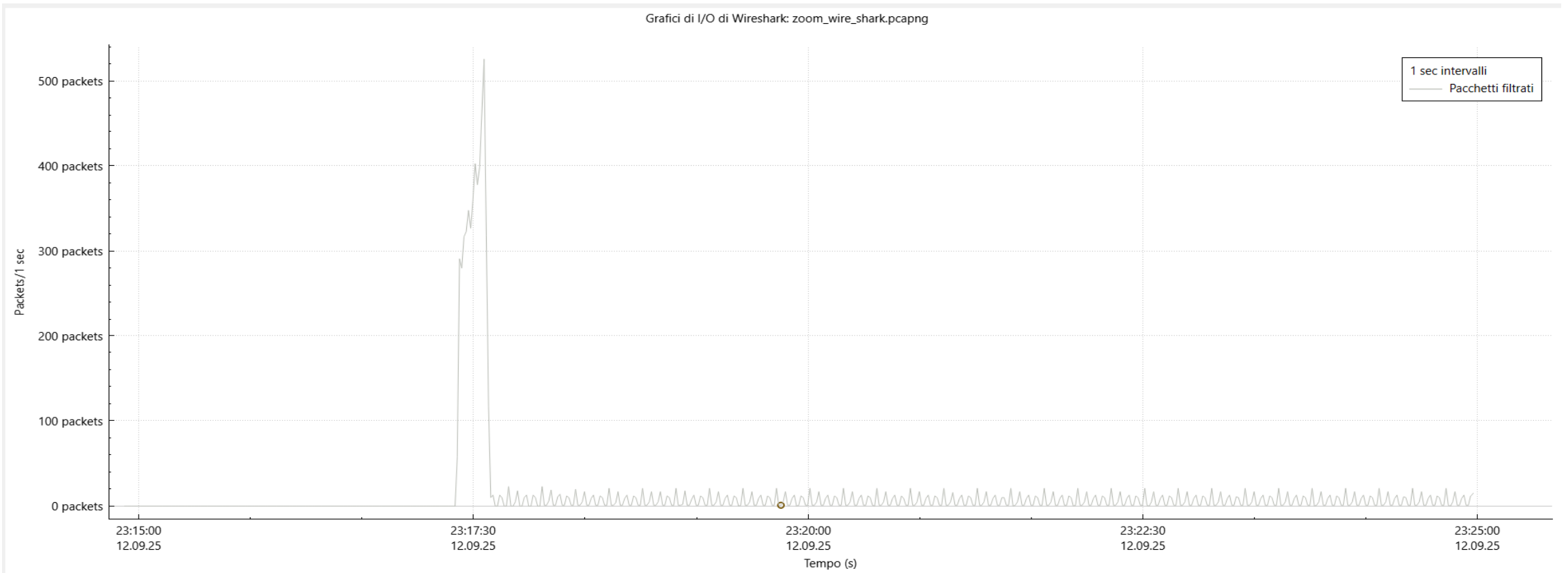
WireGuard uses UDP/51820 (not **8801**) by default

03

In Wireshark, sometimes **DTLS/SRTP** encrypted packet **headers** look like WireGuard, so incorrect labeling

04

IP **206.247.42.42** belongs to a block used by Zoom for **media servers**, not a VPN provider



I/O GRAPHICS

UDP streams are already alive even with the microphone and camera turned off



PROTOCOLS USED

In addition to the DNS, TCP, TLS, and UDP protocols, the protocols of SRTP and DTLS will be used:

- **DTLS** is essentially the version of TLS (Transport Layer Security) adapted to run on datagrams, typically UDP, rather than streams as in TCP. Manages an encrypted handshake that establishes security parameters (algorithms, keys) between the parties
- **SRTP** is a secure extension of the RTP (Real-time Transport Protocol), which is used to transport real-time media (audio/video) over the Internet. It does not itself have a complete mechanism for negotiating keys, which is why it combines with DTLS to obtain keys (model “**DTLS-SRTP**”)

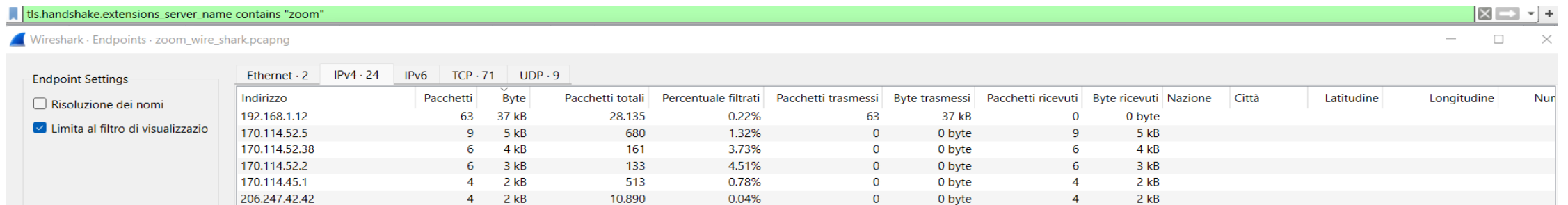
ZOOM SERVERS

Zoom uses:

- **CDN (Content Delivery Network)** and globally distributed servers: to provide static content
- **Signaling server:** to manage authentication, join/leave from meetings
- **Media server:** for audio, video and screen sharing



SERVERS CONTACTED FREQUENTLY



The image shows a Wireshark packet capture analysis. The filter bar at the top displays the filter: `tls.handshake.extensions_server_name contains "zoom"`. The packet list on the left shows several packets, with the selected packet being a TLS handshake. The packet details pane on the right shows the 'Endpoint Settings' section, which is expanded. It contains two checkboxes: 'Risoluzione dei nomi' (unchecked) and 'Limita al filtro di visualizzazio' (checked). Below the checkboxes is a table with 13 columns: 'Indirizzo', 'Pacchetti', 'Byte', 'Pacchetti totali', 'Percentuale filtrati', 'Pacchetti trasmessi', 'Byte trasmessi', 'Pacchetti ricevuti', 'Byte ricevuti', 'Nazione', 'Città', 'Latitudine', 'Longitudine', and 'Nur'. The table lists six IP addresses and their corresponding statistics.

Indirizzo	Pacchetti	Byte	Pacchetti totali	Percentuale filtrati	Pacchetti trasmessi	Byte trasmessi	Pacchetti ricevuti	Byte ricevuti	Nazione	Città	Latitudine	Longitudine	Nur
192.168.1.12	63	37 kB	28.135	0.22%	63	37 kB	0	0 byte					
170.114.52.5	9	5 kB	680	1.32%	0	0 byte	9	5 kB					
170.114.52.38	6	4 kB	161	3.73%	0	0 byte	6	4 kB					
170.114.52.2	6	3 kB	133	4.51%	0	0 byte	6	3 kB					
170.114.45.1	4	2 kB	513	0.78%	0	0 byte	4	2 kB					
206.247.42.42	4	2 kB	10.890	0.04%	0	0 byte	4	2 kB					

- The IP addresses: 170.114.45.1, 170.114.52.38, 170.114.52.2 respectively refer to the servers most used by Zoom Video Communications for the **distribution of static content** via their **CDN**
- The IP address: 206.247.42.42 is the most contacted server during the media phase through the UDP and SRTP protocols; therefore, we could consider it a **media server**
- The IP addresses: 170.114.52.5, 170.114.15.219 are the most contacted servers during the opening and closing phase of the Zoom application; therefore, we can consider them as **signaling servers**



THANKS FOR THE ATTENTION



Below is the information from the two most contacted Zoom servers:

170.114.52.5 → us05web.zoom.us

Country: United States

Region: California

City: San Jose

ISP: Cloudflare London, LLC

Organization: Zoom Video Communications, Inc

170.114.15.219 → us04zpns.zoom.us

Country: United States

Region: California

City: San Jose

ISP: Amazon.com, Inc

Organization: Zoom Video Communications, Inc